Fuzzy Triple Contrastive Learning for Hyperspectral Image Classification

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***Abstract*— Recently, contrastive learning (CL) has shown excellent performance in the hyperspectral image (HSI) classification. However, existing CL-based methods face two specific challenges: 1) multiview samples inevitably introduce ambiguity and uncertainty due to data augmentation operations. Traditional contrastive learning methods fail to effectively model these dynamic ambiguous features, resulting in a lack of robustness in the feature learning process and 2) existing CL-based methods primarily learn feature representations by pulling positive samples closer and pushing negative samples apart. However, they lack structured modeling of intraclass feature compactness and interclass feature separability. To address these challenges, we propose a fuzzy triplet contrastive learning (FTCL) method for HSI classification. For the first challenge, we propose a multiview fuzzy neighborhood learning (MFNL) module. This module effectively models the ambiguity among multiview samples through fuzzy membership calculation, multiview fuzzy weight matrix generation, and weighted feature aggregation, significantly enhancing the robustness and stability of feature representations. To tackle the second challenge, we design a triplet feature discriminative (TFD) classifier, which improves intraclass compactness by minimizing the distance between anchor samples and positive samples, while enhancing interclass separability by maximizing the distance between anchor samples and negative samples. This enables precise modeling of intraclass compactness and interclass separability. The proposed method is evaluated on four HSI datasets, and the experimental results demonstrate that the proposed method outperforms the state-of-the-art methods.**

***Index Terms*— Contrastive learning (CL), interclass similarity, intraclass diversity, multiview fuzzy neighborhood learning (MFNL).**

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I. Introduction

HYPERSPECTRAL imaging systems simultaneously acquire spectral and spatial information of terrestrial objects, yielding hyperspectral images (HSIs) for comprehensive analysis. The spectral dimension of HSIs includes tens to hundreds of continuous bands, which greatly improves the capability to differentiate ground objects. Therefore, the HSI is extensively applied in disaster monitoring, vegetation classification, and precision agriculture [1], [2], [3].

The goal of HSI classification is to differentiate the ground objects represented by each pixel in the HSI data. In the past, HSI classification tasks primarily utilized traditional machine learning algorithms. These algorithms typically involve two main processes: feature engineering and classifier classification. Feature engineering aims to preprocess the data using domain knowledge, ensuring that the extracted features can be effectively utilized by subsequent classification algorithms. For instance, dimensionality reduction techniques like principal component analysis (PCA) [4] and independent component analysis (ICA) [5] are commonly used. Popular classifiers include methods such as support vector machines (SVMs) [6], random forest (RF) [7], and K-nearest neighbors (KNNs) [8]. However, these approaches often overlook the spatial information present in HSI data, and the feature extraction process typically requires manual intervention, which can limit the enhancement of classification accuracy.

With the rapid advancement of deep learning in computer vision, end-to-end nonlinear feature learning has been widely adopted for HSI classification [9], [10]. In particular, convolutional neural networks (CNNs) [11], ranging from 1-D to 3-D architectures [12], [13], [14], have demonstrated impressive performance due to their strong feature extraction capabilities. Recent improvements, such as the S^2PNet [15] and AsyFFNet [16], further enhance the integration of spectral and spatial information. However, training deep CNNs typically relies on large amounts of labeled data, which is difficult to obtain in HSI applications due to the high cost of pixel-wise annotation. As a result, insufficient labeled samples hinder model optimization and generalization, limiting the practical deployment of large-scale deep learning models for HSI classification.

Recently, self-supervised learning [17], particularly contrastive learning (CL), has attracted significant attention for its capability to extract effective data representations without

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labeled samples. CL works by minimizing distances between positive sample pairs and maximizing those between negative pairs [18], thereby alleviating the challenge of scarce labeled data in HSI tasks. This paradigm also enhances feature transferability, as it focuses on learning generalizable representations based on relative data relationships. Numerous studies have demonstrated the efficacy of CL in HSI classification [19], [20], [21], [22], [23]. For example, Liu et al. [24] utilized dimensionality reduction to generate two spectral views for contrastive training with ResNet-50 [25], followed by SVM-based classification. Further advances, such as the HCL-Net [26] and the framework proposed by Khotimah et al. [27], have also contributed to improved HSI classification performance. However, standard CL approaches often require large numbers of negative samples, increasing data dependency and computational cost. Addressing this, Li et al. [28] adopted the Bootstrap Your Own Latent (BYOL) framework [29] for HSI classification, which eliminates the reliance on negative samples and reduces the computational burden.

Although CL methods have been effectively used in HSI classification, there exist two key challenges. First, CL relies on generating multiple augmented views of each sample, but these transformations (such as rotation, scaling, or cropping) may blur the inherent features and introduce ambiguity, especially when multiview samples from the same object exhibit noticeable differences. This can make it difficult for the model to identify crucial features, thereby affecting the robustness and generalization of learned representations. Second, traditional CL mainly encourages positive pairs (different views of the same class) to be closer and negative pairs (samples from different classes) to be farther apart, yet it lacks the explicit modeling of intraclass compactness and interclass separability. As a result, feature representations for the same class may remain scattered, while boundaries between different classes may not be sufficiently distinct, limiting the discriminative capacity of the model. Addressing these issues is vital for further improving HSI classification performance.

To address these challenges, we propose a fuzzy triplet CL (FTCL) framework. Specifically, for the ambiguity introduced by multiview augmentations, we design a multiview fuzzy neighborhood learning (MFNL) module, which models feature uncertainty through fuzzy membership computation, fuzzy weight matrix construction, and weighted feature aggregation, thereby enhancing the robustness of feature representation. Moreover, to improve both intraclass compactness and interclass separability, we introduce a triplet feature discriminative (TFD) classifier that directly optimizes anchor-positive and anchor-negative distances. This relative distance optimization effectively compensates for the lack of explicit interclass constraints in self-supervised frameworks, facilitating the construction of a more discriminative and geometrically well-structured feature space, and significantly improving generalization in complex scenarios. Our main contributions can be summarized as follows.

1) We propose a multiview asymmetric CL framework. By introducing a small number of negative samples to construct a triplet view space, the framework enhances interclass separability while maintaining intraclass compactness, effectively improving the robustness and discriminative ability of multiview feature modeling.
2) We introduce the MFNL module. This module precisely models the fuzziness between multiview samples, significantly enhancing the robustness and stability of feature representations.
3) We design the TFD classifier. By minimizing the distance between anchor and positive samples to enhance intraclass cohesion and maximizing the distance between anchor and negative samples to improve interclass separability, the module effectively models intraclass compactness and interclass discriminability.

The remainder of this article is organized as follows. Section II covers related work. Section III provides a detailed explanation of the proposed method. Section IV presents experimental results and analyses. Section V discusses the proposed method. Section VI concludes the article.

II. Related Works

A. CNN-Based Methods for HSI Classification

In recent years, deep learning approaches have enabled integrated feature extraction and classification within deep networks, substantially advancing HSI classification [30], [31]. Early works, such as Chen et al. [32] and Yue et al. [33], employed 2D-CNNs and stacked autoencoders, but were limited by insufficient exploitation of spectral information. Later, advances like dense networks [34] and 3D-CNN architectures [12] allowed for more effective modeling of spatial–spectral features, though often at the cost of increased parameters and training time. Hybrid networks, such as the HybridSN [13], combined 2-D and 3-D convolutions for richer feature extraction, yet still faced challenges in feature utilization. Other approaches integrated multisource data [35], but sometimes produced redundant features irrelevant to classification.

To further enhance feature discrimination, attention mechanisms have recently been introduced into CNN frameworks. Examples include spatial attention [36], hybrid 2-D–3-D models with 3-D attention [37], as well as residual structures to address overfitting and improve information flow [38]. Transformers and hybrid CNN–Transformer methods have also been proposed to capture long-range dependencies [39], [40]. However, the intrinsic limitation of convolutional operations—primarily processing local and regular grid data—restricts the ability of CNNs to model long-range contextual or relational information, presenting challenges in handling irregular or complex HSI data.

B. GCN-Based Methods for HSI Classification

To overcome the limitations of CNNs in capturing spatial relationships, researchers have introduced graph convolutional networks (GCNs) [41], which model pixel-level relationships as a graph, thus enabling richer spatial and contextual information extraction. Benefiting from these characteristics, GCNs have been successfully applied to HSI classification tasks [42], [43]. For example, Qin et al. [44] improved classification

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accuracy by extracting spectral–spatial features with GCNs. However, the computational cost remains high when treating each pixel as a graph node. To mitigate this, Hong et al. [45] proposed mini-batch GCN training by selecting a subset of pixels, though this limits the network's expressive capacity. Further efforts have combined GCNs with 3-D convolutions and residual connections to better capture spectral and spatial features [46], or fused graph attention with CNN attention to enhance feature representation [47]. Multihop and multiscale GCNs have also been proposed, such as the attention-based MCFN [48], which further boosts the extraction of multiscale spatial information.

The above CNN-based and GCN-based methods have been widely used in HSI classification. However, CNN-based approaches require large amounts of labeled data, which are often scarce in real-world scenarios, restricting their practical application. Similarly, GCN-based methods also face the challenge of limited labeled pixels [49], [50], and insufficient supervision makes it difficult to train GCNs with strong discriminative capabilities [51]. In contrast, our proposed method employs self-supervised CL to alleviate the dependence on labeled data. This approach learns robust feature representations without heavy reliance on annotations, substantially reducing the requirement for large-scale labeled datasets. By utilizing diverse augmentations to generate multiple views, the method improves feature discrimination by maximizing agreement between augmented samples while minimizing similarities between different samples.

C. CL-Based Methods for HSI Classification

Self-supervised learning has recently demonstrated effectiveness in mitigating the reliance on labeled data. In contrast to supervised approaches, self-supervised learning leverages unlabeled data by generating proxy supervision signals. CL, in particular, defines positive and negative pairs to encourage similar representations for positives and distinct ones for negatives, thus enabling robust feature learning. Current contrastive methods mainly include negative-sample-driven, clustering-based, asymmetric network, and redundancy-reducing approaches. Representative examples are MoCo [52] and SimCLR [18] (negative-sample-driven), and SwAV [53] (clustering-based).

Based on the advantages of CL, researchers have applied it to HSI classification. Liu et al. [54] developed a universal knowledge-embedded CL framework for supervised, unsupervised, and semi-supervised HSI classification. Braham et al. [55] introduced two positive pair mining strategies for CL on HSIs, which reduce the reliance on high-quality data augmentations and offer an effective solution for few-shot HSI classification. Additionally, Li et al. [56] proposed an end-to-end framework named guided group CL for HSI classification. This method adaptively integrates unsupervised information into supervised CL, utilizing a similarity-guided module to effectively group samples. Li et al. [57] proposed CDFS-CASCL, a cross-domain few-shot HSI classification framework that integrates cross-modal alignment and supervised CL to address prototype inaccuracy and the influence of noisy pixels, enhancing feature discriminability. Luo et al. [58] developed DCCL, a domain-collaborative CL method that integrates pseudo-labeling and cross-domain CL to enhance feature alignment for cross-domain HSI classification. Sarpong et al. [59] proposed SCTCL, a self-correlation framework with triple CL (cluster-, instance-, structure-level) to handle noisy labels and improve feature reliability.

Compared to existing CL methods for HSI classification, FTCL demonstrates clear advantages in two main areas. First, to address the dynamic ambiguity caused by multiview augmentations, FTCL integrates an MFNL module that uses fuzzy membership and weighted feature aggregation to effectively model feature uncertainty, thereby enhancing the robustness and stability of representations. Second, FTCL employs a TFD classifier, which enforces intraclass compactness and interclass separability, not only reducing distances within the same class but also maximizing margins between different classes. Together, these components address the challenges of feature uncertainty and structured modeling, resulting in improved discriminability, classification accuracy, and generalization in HSI classification.

III. Proposed Method

A. Overview of the Framework

The overall structure of the proposed FTCL is illustrated in Fig. 1. FTCL utilizes self-supervised pretraining followed by classification. During pretraining, multiview perspectives are constructed via data augmentation. The MFNL module extracts fuzzy-enhanced spectral features via fuzzy membership-weighted aggregation. These features are projected to the TFD classifier, which enhances class separation by minimizing anchor-positive distances and maximizing anchor-negative distances. Finally, the classification uses the pretrained encoder, a fully connected layer, and a Softmax layer for output.

B. Data Preprocessing and Augmentation

To prepare HSI data for model training, we adopted a dimensionality reduction approach combining PCA and maximum noise fraction (MNF). PCA reduces spectral dimensions from c to q, preserving significant variance, while MNF enhances the signal-to-noise ratio (SNR) by suppressing irrelevant noise. In our experiments, the value of q was empirically set to 30, which retains over 99% of the cumulative explained variance. This two-step method reduces redundancy while retaining essential features and lowering computational costs. The transformation is expressed as

$$\mathbf{X}_{\text{reduced}} = \mathbf{W}_{\text{MNF}} \cdot (\mathbf{W}_{\text{PCA}} \cdot \mathbf{X}) \tag{1}$$

where $\mathbf{W}_{\text{PCA}}$ and $\mathbf{W}_{\text{MNF}}$ are the transformation matrices for PCA and MNF.

A sliding window was then applied to divide the image into $d \times d$ subblocks, integrating spectral and spatial features. This enriches each pixel vector X_i with local contextual information, enabling the model to better capture spatial relationships. To improve robustness in real-world scenarios, random Gaussian noise was added during data augmentation.

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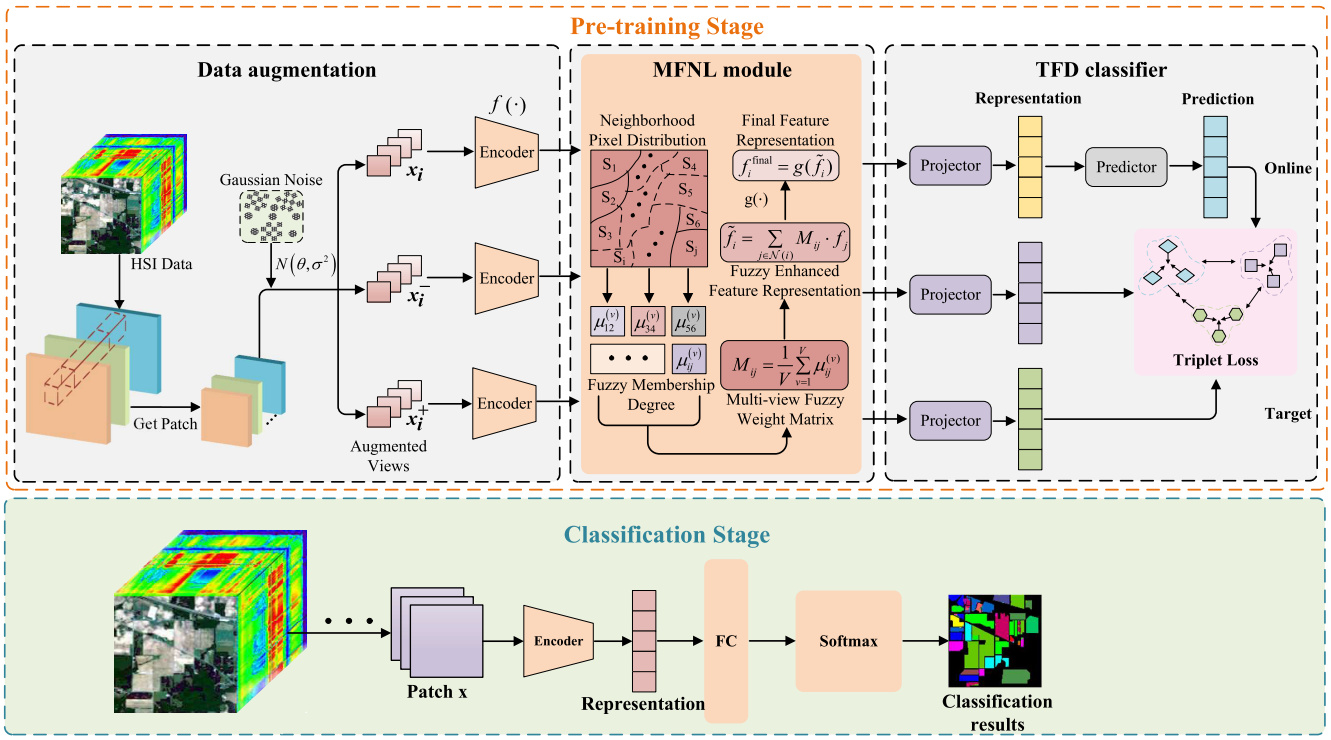


Fig. 1. Framework of FTCL. In the self-supervised pretraining stage, multiview samples are generated via data augmentation and processed by the online and target networks to extract features. The MFNL module captures spectral features and computes fuzzy membership, while the TFD classifier enhances feature discrimination. During the classification stage, the pretrained encoder is fixed, and a fully connected layer with a Softmax layer is added to produce classification results.

Unlike noise suppression during dimensionality reduction, this step simulates environmental interference, enhancing generalization. The augmentation formula is

$$\mathbf{X}' = \alpha \cdot \mathbf{x} + \beta \cdot \mathbf{N} \tag{2}$$

where $\mathbf{N}$ is Gaussian noise, α is a random parameter within [0,1], and β is set to 1/25. This preprocessing pipeline ensures high-quality feature extraction and improves adaptability to complex environments.

Positive samples are generated by applying Gaussian noise-based augmentation to the same original pixel, producing multiple views that preserve semantic consistency while introducing controlled perturbations. Negative samples are formed by applying the same augmentation independently to different pixels within the training batch, ensuring semantic discrepancy. This random noise augmentation strategy improves model robustness and generalization without the need for complex hard negative mining. It is consistent with established CL practices, balancing training stability, computational efficiency, and reproducibility. In this framework, three augmented views, $v = t(x)$, $v^+ = t^+(x)$, and $v^- = t^-(x)$, are generated. The online network produces feature representations y_θ based on v and y'_θ based on v^-, while the target network outputs representation y_ξ from v^+. Feature extraction is formally defined as

$$y_\theta \triangleq f_\theta(v) \tag{3}$$

$$y'_\theta \triangleq f_\theta\left(v^-\right) \tag{4}$$

$$y_\xi \triangleq f_\xi\left(v^+\right) \tag{5}$$

where f_θ is the online network encoder, θ is the online network parameter, f_ξ is the target network encoder, and ξ is the target network parameter. The target network has the same architecture as the online network but uses different network parameter settings.

We propose a feature extraction network tailored for HSIs, aiming to fully exploit joint spectral–spatial features and enhance discriminative capability. The detailed structure of the extractor is shown in Fig. 2. The extractor adopts a progressive layered convolutional architecture, consisting of both 3-D and 2-D convolutional modules to capture multiscale spatial and spectral information. Initially, the input HSI undergoes a series of 3-D convolutions and downsampling, where key spectral bands and spatial regions are dynamically enhanced via Spectral Attention and Spatial Attention mechanisms. Shallow detail features and deep semantic representations are then fused through skip connections and further refined by a 2-D convolutional module. Finally, the high-dimensional features are flattened and mapped to fully connected layers to generate compact feature embeddings for classification or other downstream tasks. This structure offers significant advantages in deep feature modeling, multiscale processing, and the enhancement of hyperspectral data representations.

The target network provides the regression objective for online network training, where parameter ξ is the exponential moving average of online parameter θ. Specifically, given a target attenuation rate $\tau \in (0, 1)$, after each training step, perform the following updates:

$$\xi \leftarrow \tau\xi + (1 - \tau)\theta. \tag{6}$$

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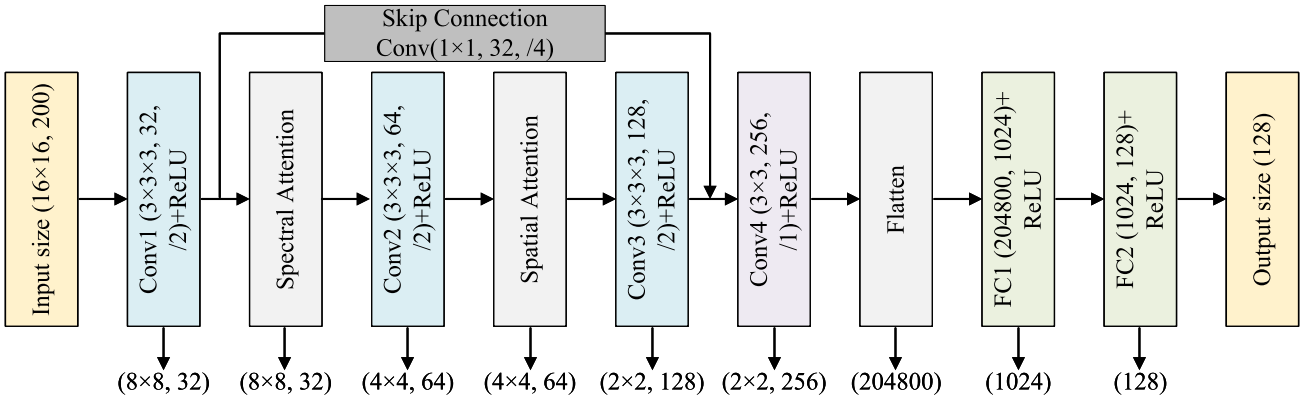


Fig. 2. Structure of the hyperspectral feature extraction network. Batch Normalization and ReLU activations after each convolutional layer are omitted for clarity. The attention mechanism adopts a two-stage structure: 1) spectral attention generates channel weights via global spatial pooling, fully connected layers, and a sigmoid activation and 2) spatial attention obtains spatial weights by concatenating channel-wise pooled features, followed by convolution and a sigmoid activation. The skip connection applies a 1×1 convolution for channel alignment before concatenation.

FTCL utilizes an asymmetric network structure and a momentum update mechanism to avoid convergence to local optima. The online-target network design promotes diverse feature representations, while the momentum update stabilizes parameter changes, preventing solution collapse. Together, these mechanisms ensure robust feature learning, improved generalization, and stable convergence during training.

C. MFNL Module

Due to various factors such as imaging distance, illumination, land cover, and environmental conditions, objects within the same class in HSI data may exhibit certain differences, while objects from different classes often share similar visual characteristics. This can result in classification uncertainties, including intraclass heterogeneity and interclass ambiguity. To address this issue, we propose a specifically designed MFNL module for processing the feature representations of three different view samples. Unlike traditional FNL modules, this module is optimized for multiview samples generated through data augmentation. It utilizes fuzzy membership degree calculation and a multiview fuzzy weight matrix to perform weighted aggregation of features, thereby enhancing the robustness and discriminative ability of feature representations. The structure of MFNL is shown in Fig. 3.

1) Fuzzy Membership Matrix: Fuzzy membership serves as the cornerstone of the MFNL module, quantifying the similarity between a pixel and its neighboring pixels. For a pixel x_i and its neighboring pixel x_j (where the neighborhood is determined by a window size of $W \times W$), the fuzzy membership $\mu_{ij}^{(v)}$ in a specific view v is defined as

$$\mu_{ij}^{(v)} = \exp\left(-\frac{\|f_i^{(v)} - f_j^{(v)}\|^2}{\sigma^2}\right) \tag{7}$$

where $f_i^{(v)}$ and $f_j^{(v)}$ represent the d-dimensional feature vectors of pixel x_i and its neighboring pixel x_j in view v, respectively. The term $\|f_i^{(v)} - f_j^{(v)}\|^2 = \sum_{k=1}^{d}(f_i^{(v)}[k] - f_j^{(v)}[k])^2$ denotes the Euclidean distance between the feature vectors, and σ is the smoothing parameter that controls the rate of decay for the membership value. The fuzzy membership values range between $[0, 1]$, with higher values indicating greater similarity between x_i and x_j. To facilitate computation, the fuzzy membership matrix $\mu^{(v)} \in \mathbb{R}^{N\times N}$ can be expressed as

$$\mu_{ij}^{(v)} = \exp\left(-\frac{\|f_i^{(v)} - f_j^{(v)}\|^2}{\sigma^2}\right) \quad \forall i, j \in \{1, 2, \ldots, N\} \tag{8}$$

where N represents the total number of pixels. The computation of fuzzy membership captures the local similarity relationships between a pixel and its neighbors, thereby incorporating spatial contextual information into the feature representation. This similarity modeling provides a foundation for subsequent feature fusion and helps mitigate the impact of intraclass heterogeneity on feature consistency.

2) Multiview Fuzzy Weight Matrix: Based on multiview data, the MFNL module further integrates the fuzzy membership values across all views to generate a multiview fuzzy weight matrix M_{ij}. Specifically, M_{ij} is computed as

$$M_{ij} = \frac{1}{V}\sum_{v=1}^{V}\mu_{ij}^{(v)} \tag{9}$$

where V denotes the total number of views (in this case, $V = 3$), and $\mu_{ij}^{(v)}$ represents the fuzzy membership in view v. By representing the fuzzy weight matrix for all pixels as $M \in \mathbb{R}^{N\times N}$, the matrix can be expressed as

$$\mathbf{M} = \frac{1}{V}\sum_{v=1}^{V}\mu^{(v)}. \tag{10}$$

By fusing fuzzy membership information across different views, the multiview fuzzy weight matrix comprehensively captures the fuzzy relationships between a pixel and its neighbors. This multiview fusion mechanism effectively alleviates potential biases in individual views and more comprehensively captures the spectral–spatial characteristics of pixels, thereby significantly improving the robustness and discriminative capability of feature representations.

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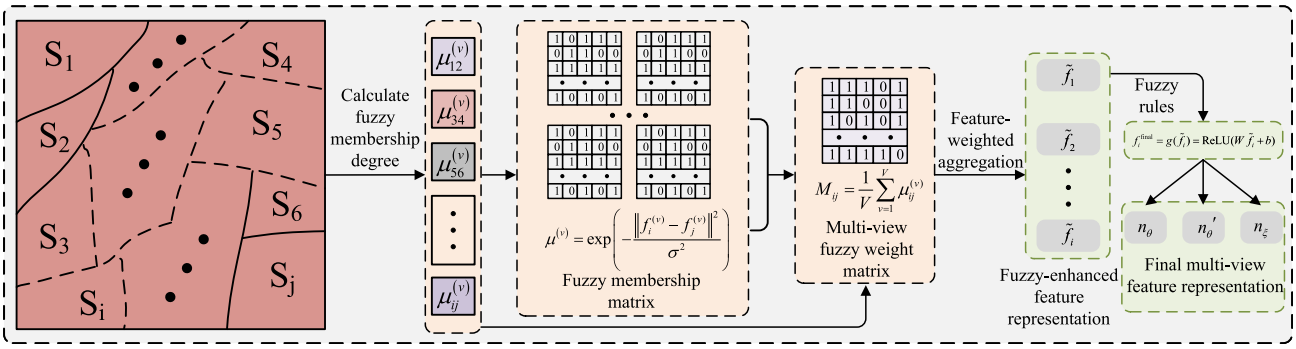


Fig. 3. Structure of MFNL. This module calculates fuzzy membership degrees from input data to construct a multiview fuzzy weight matrix. Feature-weighted aggregation uses this matrix to produce a fuzzy-enhanced feature representation. Fuzzy rules then refine these features, generating a final multiview feature representation for analysis and prediction.

3) Feature Weighted Aggregation: After generating the multiview fuzzy weight matrix M_{ij}, the MFNL module performs weighted aggregation of the features of neighboring pixels to obtain the fuzzy-enhanced feature representation $\tilde{f}_i$. The computation is defined as

$$\tilde{f}_i = \sum_{j \in \mathcal{N}(i)} M_{ij} \cdot f_j \tag{11}$$

where $\mathcal{N}(i)$ denotes the set of neighboring pixels of x_i, f_j represents the original feature vector of the neighboring pixel x_j, and M_{ij} is the corresponding weight in the multiview fuzzy weight matrix. Representing the feature vectors for all pixels as a matrix $\mathbf{F} \in \mathbb{R}^{N \times d}$, the fuzzy-enhanced feature representation can be expressed as

$$\tilde{\mathbf{F}} = \mathbf{M} \cdot \mathbf{F}. \tag{12}$$

This process effectively reduces the impact of noisy and irrelevant pixels. Since noisy pixels typically exhibit low similarity with their neighbors, their fuzzy membership values are small, resulting in minimal contribution during the weighted aggregation. Simultaneously, the fuzzy weight matrix emphasizes neighbors with similar features to the target pixel while suppressing the influence of unrelated pixels, thereby improving the purity and consistency of the feature representation.

4) Application of Fuzzy Rules: On the basis of the fuzzy-enhanced feature representation $\tilde{f}_i$, the MFNL module applies nonlinear fuzzy rules to generate the final feature representation f_i^{final}. The computation is defined as

$$f_i^{\text{final}} = g\left(\tilde{f}_i\right) \tag{13}$$

where the nonlinear transformation function $g(\cdot)$ is a learnable module implemented as an MLP, consisting of several fully connected layers with ReLU activation functions. The parameters of this module are jointly optimized with the rest of the model via backpropagation during training. This design allows the fuzzy rule function to adaptively learn complex nonlinear mappings from the fuzzy-enhanced features, effectively extracting more discriminative patterns while suppressing irrelevant variations. For all pixels, the final feature representation can be expressed as

$$\mathbf{F}^{\text{final}} = g\left(\tilde{\mathbf{F}}\right). \tag{14}$$

The application of fuzzy rules further extracts discriminative information from the features, making the final feature representation more distinctive. By leveraging nonlinear transformations, the module captures more complex patterns within the features, thereby improving the classifier's ability to distinguish between categories. The three features of HSI processed by the MFNL module can be represented as n_θ, n'_θ, and n_ξ

$$n_\theta \triangleq w_i(y_\theta) \tag{15}$$

$$n'_\theta \triangleq w_i\left(y'_\theta\right) \tag{16}$$

$$n_\xi \triangleq w_i\left(y_\xi\right) \tag{17}$$

where $w_i(\cdot)$ is the fuzzy weighted aggregation process of the MFNL module.

D. TFD Classifier

We use an asymmetric online target network to process fuzzified samples. The online network includes an encoder, a projection head, and a predictor, while the target network consists of an encoder and a projection head. The projection head applies a nonlinear transformation to encoder features, mapping them to the triplet loss feature space. The nonlinear two-layer projection head can be expressed as

$$g(f(x)) = W^{(2)}\left(\sigma\left(W^{(1)}(f(x))\right)\right) \tag{18}$$

where $W^{(1)}$ and $W^{(2)}$ are the weights of the fully connected layers, and σ is the ReLU activation function. The projection heads of the online and target networks output three types of feature representations

$$z_\theta \triangleq g_\theta(n_\theta) \tag{19}$$

$$z'_\theta \triangleq g_\theta\left(n'_\theta\right) \tag{20}$$

$$z_\xi \triangleq g_\xi\left(n_\xi\right) \tag{21}$$

where g_θ and g_ξ are the mapping functions of the online and target networks, with parameters θ and ξ, respectively. The

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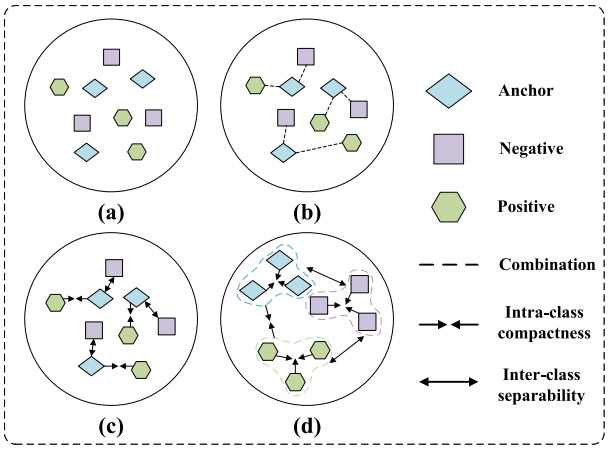


Fig. 4. Triplet loss. (a) In the initial sample feature distribution, the triplet loss function is applied. (b) Anchor, positive, and negative samples are selected to form triplets. (c) Distances between the anchor and positive samples, and between the anchor and negative samples, are calculated. This process brings similar features closer together while pushing dissimilar features further apart. (d) After multiple iterations of optimization, convergence is achieved. This results in samples of the same class being sufficiently clustered in the feature space, while samples of different classes are adequately separated.

anchor sample representation z_θ is further processed by the online network predictor $q_\theta(z_\theta)$. This asymmetric structure and distinct parameter updates effectively prevent model collapse and enhance performance.

The InfoNCE loss performs well in CL but struggles in high-dimensional spaces with significant intraclass heterogeneity or interclass ambiguity. Its pairwise design limits its ability to model multiview data. To address this, we design the TFD classifier, which optimizes relative distances by ensuring the anchor-positive distance is smaller than the anchor-negative distance. This enhances intraclass compactness, and interclass separability, and better integrates multiview feature information compared to InfoNCE. The process involves creating triplets with an anchor sample (A), a negative sample (N), and a positive sample (P). The construction and optimization process of the triplets is illustrated in Fig. 4.

The TFD classifier enhances the self-supervised learning framework by explicitly optimizing the feature space, addressing issues like loose intraclass distributions and poor interclass separability. Pulling similar samples closer and pushing dissimilar ones apart creates a more structured and discriminative feature space, improving the model's generalization across tasks and datasets.

The goal of the TFD classifier is to ensure that the distance between the anchor sample A and the positive sample P is less than the distance between the anchor sample A and the negative sample N, with at least one preset margin. Therefore, we can obtain

$$\left\|f\left(x_i^A\right)-f\left(x_i^p\right)\right\|^2+m<\left\|f\left(x_i^A\right)-f\left(x_i^N\right)\right\|^2. \tag{22}$$

Calculate the distance between the anchor sample A and the positive sample P, denoted as $\|f(x_i^A)-f(x_i^p)\|$. Calculate the distance between the anchor sample A and the negative sample N, denoted as $\|f(x_i^A)-f(x_i^N)\|$. Then, compute the difference in distance between the two previously calculated distances and add the margin m, as shown in (22). In (22), $f(x)$ represents the feature representation of the sample x obtained through the BYOL network and m is the preset margin. We designed a triplet loss function for the TFD classifier, as shown in the following formula:

$$L_{\text{triplet}}=\max\Big(0,\left\|f\left(x_i^A\right)-f\left(x_i^p\right)\right\|^2 -\left\|f\left(x_i^A\right)-f\left(x_i^N\right)\right\|^2+m\Big) \tag{23}$$

where $\|\cdot\|^2$ represents the squared Euclidean distance and m is a positive hyperparameter, referred to as the "margin," which ensures that there is sufficient separation between the anchor sample and the negative sample.

The triplet loss function continuously optimizes the FTCL model parameters through backpropagation to ensure that the distance between the anchor sample and the negative sample is greater than the distance between the anchor sample and the positive sample, with the difference between the two distances being at least greater than the preset margin m. The objective is to achieve a target loss of zero, which can be formulated as follows:

$$\left\|f\left(x_i^A\right)-f\left(x_i^N\right)\right\|^2-\left\|f\left(x_i^A\right)-f\left(x_i^p\right)\right\|^2>m. \tag{24}$$

Based on the aforementioned analysis, by leveraging the MFNL module and the TFD classifier, we can train an exceptional encoder $f(\cdot)$. Subsequently, we select HSI classification as a downstream task to evaluate the generalization capability of the encoder. In this downstream task, we keep the encoder parameters fixed as learned during the pretraining phase, remove the original projection head, and introduce a new fully connected layer and a Softmax layer as the classifier. The definition of Softmax is as follows:

$$\text{softmax}(z_i)=\frac{\exp(z_i)}{\sum_{j=1}^{K}\exp\left(z_j\right)} \tag{25}$$

where z_i and z_j are real-valued vectors. The Softmax function converts a vector into a probability distribution.

In summary, the flow of FTCL is as follows.

IV. Experiments

A. Datasets

1) Indian Pines Dataset: The Indian Pines (IP) dataset consists of HSIs captured from the Indian Pines test site in Indiana, USA. This dataset includes 145 × 145 pixels and 200 spectral bands, covering 16 distinct land cover types.

2) Pavia University Dataset: The Pavia University (PU) dataset consists of HSIs captured from the PU area in Pavia, Italy. This dataset includes 103 spectral bands and covers an area of 610 × 340 pixels, representing 9 distinct land-cover types.

3) WHU_Hi_LongKou Dataset: The WHU_Hi_LongKou (LongKou) dataset contains HSIs captured from the Longkou region in Shandong Province, China. This dataset features a spatial resolution of 550 × 400 pixels and comprises 270 spectral bands, representing 9 distinct crop types.

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Algorithm 1 FTCL for HSI Classification

Input: HSI data X

Output: encoder $f(\cdot)$

Obtaining reduced dimensional HSI by PCA and MNF;
Extract local spatial feature block x_i of the image through the sliding window of $d \times d$;
for epochs **do**
  for batches **do**
    $X' \leftarrow \alpha \cdot x + \beta \cdot N$ ▷ Data augmentation
    Generate feature representations y_θ, y'_θ, and y_ξ for three augmented views using formulas 3, 4, and 5;
  for i=1 to k **do**
    $\mu_{ij}^{(v)} = \exp(-\frac{\|f_i^{(v)} - f_j^{(v)}\|^2}{\sigma^2})$ ▷ Calculate the fuzzy affiliation under a particular view v
  for i=1 to k **do**
    $M_{ij} = \frac{1}{V}\sum_{v=1}^{V} \mu_{ij}^{(v)}$ ▷ Generate multiview fuzzy weight matrix
    Input feature map F_n, process it with the final fuzzy weight matrix M, and output the fuzzy enhanced feature map F'_n;
  Output the mapping expressions z_θ, z'_θ, and z_ξ for three feature representations using formulas 19, 20, and 21;
  Construct a triplet $[z_\theta, z'_\theta, z_\xi]$;
  $L_1 \leftarrow \|f(z_\theta) - f(z_\xi)\|^2$ ▷ Calculate the squared Euclidean distance between anchor samples and positive samples
  $L_2 \leftarrow \|f(z_\theta) - f(z'_\theta)\|^2$ ▷ Calculate the squared Euclidean distance between anchor samples and negative samples
  $L_{\text{triplet}} \leftarrow \max(0, L_1 - L_2 + m)$ ▷ Calculate triplet loss

4) WHU_Hi_HongHu Dataset: The WHU_Hi_HanChuan (HanChuan) dataset contains HSIs acquired from the Hanchuan region in Hubei Province, China. This dataset has a spatial size of 1217 × 303 pixels, with 270 spectral bands, and represents 16 distinct land-cover categories.

In this study, ten samples per class were randomly selected as the training set, with the remaining samples used for testing. The choice of a limited number of training samples per class was deliberate, aiming to reflect the common real-world constraint of scarce labeled data in HSI classification tasks. This experimental design enables a rigorous assessment of the model's capability to generalize and maintain robustness under small-sample conditions. The specific distributions of training and testing samples for the four datasets are detailed in Table I.

B. Experimental Settings

1) Comparison With Other Models: To evaluate the performance of FTCL, we selected eight representative methods in the field of HSI classification for comparative experiments.

These methods include: 2D-CNN [3], a traditional model utilizing CNN; Spectral Former [60], a network based on the Transformer architecture; SSGRN [61], a semi-supervised learning algorithm employing GCNs. Additionally, we considered DMVL+SVM [23] and S3Net [62], both established methods in the domain of two-stage CL. Furthermore, to facilitate a meaningful comparison with current mainstream CL frameworks, we included GGCL [56], SSCL [63], and ConVaT [64] in our comparative experiments. The hyperparameters for these comparison methods were adjusted based on optimal configurations or recommended settings mentioned in their respective papers to ensure the fairness and comparability of the experimental results.

1) *2D-CNN [3]:* 2D-CNN uses filters to extract features and create multiscale feature maps through convolution operations, capturing local spatial structure information.

2) *Spectral Former [60]:* a Transformer-based approach, focuses on understanding the global spectral space and leveraging local spectral–spatial information to improve classification accuracy.

3) *SSGRN [61]:* SSGRN utilizes the GCN to integrate spectral and spatial information for capturing complex feature relationships.

4) *DMVL+SVM [23]:* DMVL+SVM method involves learning high-dimensional feature representations from multiple perspectives and using an SVM classifier for effective class separation.

5) *S3Net [62]:* S3Net consists of two subnetworks with shared weights for processing spectral features and spatial neighborhood information. It utilizes CL to identify classes by learning sample similarities, allowing effective classification with limited samples.

6) *GGCL [56]:* GGCL is an unsupervised learning method that improves feature representation through CL of feature groups, combining CL and adaptive CL to enhance the classification performance of HSI.

7) *SSCL [63]:* SSCL utilizes a large number of unlabeled samples and data augmentation to create positive and negative sample pairs for CL during network pretraining, followed by fine-tuning with a small number of labeled samples.

8) *ConVaT [64]:* ConVaT combines variational autoencoder (VAE) and Transformer architecture to capture both spectral and spatial features of HSIs. To improve classification accuracy with limited labeled samples, ConVaT introduces a momentum contrast (MoCo) mechanism.

2) Evaluation Metrics: In the experiments, we assessed the classification performance of the model using overall accuracy (OA), average accuracy (AA), and the Kappa coefficient as evaluation metrics. To ensure the reliability of the results, each experiment was repeated ten times, and the final result was obtained by averaging these ten results.

3) Hyperparameter Settings and Implementation Details: The smoothing parameter σ was set to 0.3 to accurately model local pixel similarity. The initial learning rate was 0.001, and training was performed for 100 epochs using the Adam optimizer. Key hyperparameters, including patch size and batch size, were systematically evaluated in the parameter study section. Experiments were conducted on an NVIDIA GeForce RTX 3090 GPU and Intel Xeon Gold 6234 CPU. The implementation utilized Python 3.8, PyTorch 1.9.1, and CUDA 11.7.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES

No.	Indian Pines			Pavia University			WHU_Hi_LongKou			WHU_Hi_HanChuan		
	Class	Train	Test	Class	Train	Test	Class	Train	Test	Class	Train	Test
C1	Alfalfa	10	36	Asphalt	10	6621	Water	10	34501	Water	10	44725
C2	Corn-notill	10	1418	Meadows	10	18639	Paddy Field	10	8364	Paddy Field	10	22743
C3	Corn-mintill	10	820	Gravel	10	2089	Wheat	10	3021	Wheat	10	10277
C4	Corn	10	227	Trees	10	3054	Corn	10	63202	Corn	10	5343
C5	Pasture	10	473	Metalsheets	10	1335	Sugar Beet	10	4141	Soybean	10	1190
C6	Grass-tress	10	720	Baresoil	10	5019	Bare Soil	10	11844	Sugar Beet	10	4523
C7	Pasture-mowed	10	18	Bitumen	10	1320	Building	10	67046	Sunflower	10	5893
C8	Hay-windrowed	10	468	Bricks	10	3672	Road	10	7114	Cotton	10	17968
C9	Oats	10	10	Shadow	10	937	Forest	10	5219	Bare Soil	10	9459
C10	Soybean-notill	10	962	–	–	–	–	–	–	Building	10	10506
C11	Soybean-mintill	10	2445	–	–	–	–	–	–	Road	10	16901
C12	Soybean-clean	10	583	–	–	–	–	–	–	Woody Grassland	10	3669
C13	Wheat	10	195	–	–	–	–	–	–	Shrubland	10	9106
C14	Woods	10	1255	–	–	–	–	–	–	Orchard	10	18550
C15	Building-Grass	10	376	–	–	–	–	–	–	Grassland	10	1126
C16	Stone-Steel-towers	10	83	–	–	–	–	–	–	Shadow	10	75391
	Total	160	10089	Total	90	42686	Total	90	204452	Total	160	257370

C. Classification Results of Different Models

1) Detailed Results Analysis: Tables II–V display the classification results of various methods. The results on the four datasets indicate that while the eight advanced comparison methods show strong competitiveness, FTCL outperforms them in terms of OA, AA, and Kappa.

On the IP dataset, FTCL achieved OA, AA, and Kappa coefficient scores of 96.60%, 94.22%, and 96.12%, respectively, surpassing the second-best ConVat method by 0.65%, 1.63%, and 0.74%. In seven categories, FTCL demonstrated significant performance advantages, achieving 100% classification accuracy in C1, C7, and C13, and 98.53% in C14. On the PU dataset, FTCL achieved OA, AA, and Kappa coefficient scores of 95.55%, 93.03%, and 94.07%, respectively, surpassing the ConVat method by 1.09%, 2.75%, and 1.42%. In C1, C2, C4, C8, and C9, the accuracy of FTCL significantly outperformed other methods, achieving 96.86%, 98.22%, 96.15%, 89.55%, and 94.09%, respectively. On the LongKou dataset, FTCL achieved OA, AA, and Kappa coefficient scores of 97.35%, 97.59%, and 97.05%, respectively, surpassing the ConVat method by 0.95%, 0.79%, and 1.05%. Additionally, FTCL achieved 100% classification accuracy in C2, C3, C7, and C9. On the HanChuan dataset, FTCL achieved OA, AA, and Kappa coefficient scores of 96.99%, 97.54%, and 95.41%, respectively, surpassing the second-best ConVaT method by 2.44%, 1.62%, and 0.86%, respectively. In seven categories, FTCL demonstrated significant performance advantages, achieving 100% classification accuracy in C1, C7, C8, C9, and C13, and 99.95% in C14.

2D-CNN and Spectral Former exhibited good classification accuracy in certain categories across the three datasets. 2D-CNN uses multiple convolutional layers and pooling layers to extract the local spatial structure information of HSI layer by layer. Spectral Former relies on self-attention mechanisms to capture long-range dependencies. However, HSI contains rich spatial and spectral information and neither 2D-CNN nor Spectral Former can maintain a comprehensive understanding of the global spectral–spatial information. Consequently, their classification performance is inferior to that of the SSGRN and DMVL+SVM methods. SSGRN and DMVL+SVM can utilize spectral and spatial information for global reasoning. SSGRN improves classification accuracy by constructing a spectral–spatial global graph, which simultaneously captures spectral and spatial information. DMVL+SVM employs multiview learning to integrate multimodal features from different perspectives, comprehensively capturing the complex spatial and spectral information of HSI. However, in small-sample tasks, both SSGRN and DMVL+SVM methods exhibit the drawback of requiring a large number of labeled samples. S3Net utilizes a Siamese network structure to extract spectral and spatial features, enabling effective feature learning and similarity assessment with a small number of training samples, thereby improving classification accuracy. GGCL, SSCL, and ConVaT, as CL-based methods, exhibit superior classification performance. GGCL sets up guiding groups to enable the model to more effectively separate information from different classes in the feature space while aggregating information from the same class. This approach leverages unsupervised information to improve classification accuracy. SSCL combines the advantages of the CNN and CL. It extracts preliminary features through convolutional layers and then uses a CL module for feature optimization, significantly improving classification accuracy. ConVaT combines the advantages of momentum CL and variational generative models, exhibiting higher classification accuracy and robustness.

Although each method has its unique advantages, FTCL demonstrates the best performance in classification tasks. Its superior performance is primarily attributed to the following three key factors. These components address the challenges of ambiguity in multiview samples and the need for compact and separable feature representations, enabling FTCL to surpass existing methods.

1) *Efficient Utilization of Unlabeled Data:* FTCL employs a multiview asymmetric network to generate augmented samples and leverages self-supervised learning to iteratively extract and enhance spectral–spatial features. This approach maximizes

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TABLE II

CLASSIFICATION EXPERIMENT OF RESULTS ON THE IP (%). BOLD INDICATES THE BEST RESULTS. STANDARD DEVIATIONS ARE APPENDED

Class	2D-CNN	Spectral Former	SSGRN	DMVL + SVM	S3Net	GGCL	SSCL	ConVaT	FTCL
1	75.00 ± 3.12	73.53 ± 4.28	72.22 ± 3.89	79.41 ± 2.76	74.47 ± 3.20	75.00 ± 3.04	**100.0 ± 0.00**	93.18 ± 1.05	**100.0 ± 0.00**
2	45.65 ± 5.87	40.14 ± 6.20	76.28 ± 3.35	80.79 ± 2.63	86.93 ± 2.73	76.54 ± 3.12	94.08 ± 1.02	93.85 ± 1.10	**97.68 ± 0.75**
3	53.85 ± 5.40	53.42 ± 5.90	73.41 ± 4.01	94.81 ± 1.26	80.07 ± 2.38	94.71 ± 1.22	95.54 ± 1.12	95.96 ± 1.07	**97.09 ± 0.72**
4	43.19 ± 6.04	69.70 ± 4.12	87.88 ± 2.86	72.32 ± 4.45	76.43 ± 3.48	65.77 ± 5.01	88.51 ± 2.98	**100.0 ± 0.00**	97.69 ± 0.63
5	75.48 ± 3.22	88.42 ± 2.73	83.06 ± 3.01	75.00 ± 3.09	78.57 ± 3.56	75.20 ± 3.10	93.85 ± 1.05	**94.88 ± 1.11**	92.06 ± 1.30
6	64.71 ± 4.51	85.34 ± 2.92	80.91 ± 3.30	67.81 ± 4.23	81.30 ± 3.14	94.29 ± 1.30	94.94 ± 1.17	**98.60 ± 0.50**	97.61 ± 0.61
7	32.73 ± 7.05	15.60 ± 8.21	54.05 ± 6.43	**100.0 ± 0.00**	**100.0 ± 0.00**	**100.0 ± 0.00**	61.90 ± 5.87	**100.0 ± 0.00**	**100.0 ± 0.00**
8	67.77 ± 4.12	49.22 ± 6.88	78.62 ± 3.75	94.90 ± 1.18	91.61 ± 1.60	93.81 ± 1.22	96.72 ± 1.07	**97.95 ± 0.95**	97.06 ± 0.97
9	22.22 ± 8.20	13.45 ± 9.10	27.03 ± 7.89	30.65 ± 7.45	60.00 ± 5.50	78.95 ± 4.11	85.71 ± 3.79	**90.00 ± 2.67**	75.00 ± 4.92
10	63.80 ± 4.53	41.22 ± 6.34	81.84 ± 3.25	83.83 ± 2.87	80.43 ± 2.98	88.07 ± 2.40	89.34 ± 2.19	**97.67 ± 0.65**	95.20 ± 0.94
11	82.06 ± 3.40	78.97 ± 3.25	86.37 ± 2.73	88.68 ± 2.37	93.27 ± 1.88	97.10 ± 1.05	95.67 ± 1.14	**98.49 ± 0.60**	97.14 ± 0.59
12	51.78 ± 5.45	38.78 ± 6.32	55.71 ± 5.92	72.09 ± 4.87	82.30 ± 3.87	83.47 ± 3.74	77.91 ± 3.02	92.87 ± 1.21	**95.80 ± 0.69**
13	80.65 ± 3.35	37.14 ± 6.66	68.80 ± 4.02	85.10 ± 2.54	81.16 ± 3.56	93.57 ± 1.93	94.32 ± 1.22	81.94 ± 3.45	**100.0 ± 0.00**
14	83.38 ± 3.03	66.60 ± 4.09	94.99 ± 1.35	92.12 ± 1.66	82.34 ± 3.69	94.51 ± 1.88	98.04 ± 0.95	95.95 ± 0.73	**98.53 ± 0.37**
15	81.46 ± 3.47	63.28 ± 5.75	80.56 ± 3.31	75.00 ± 3.01	72.83 ± 4.10	68.69 ± 4.76	**97.01 ± 1.08**	89.32 ± 1.49	96.67 ± 0.88
16	54.55 ± 5.66	12.12 ± 7.88	46.25 ± 6.73	87.50 ± 2.92	55.70 ± 5.51	70.00 ± 5.34	94.81 ± 1.10	**97.14 ± 0.53**	70.54 ± 4.87
OA	65.98 ± 3.54	57.99 ± 4.12	79.93 ± 2.95	83.41 ± 2.12	84.55 ± 1.91	87.26 ± 1.56	93.61 ± 1.07	95.95 ± 0.76	**96.60 ± 0.55**
AA	59.25 ± 4.07	51.34 ± 4.65	71.43 ± 3.58	74.84 ± 3.35	76.05 ± 2.97	85.08 ± 2.04	87.90 ± 1.73	92.59 ± 1.12	**94.22 ± 0.69**
Kappa	61.21 ± 3.82	51.89 ± 4.21	77.16 ± 3.41	80.99 ± 2.54	82.39 ± 2.22	85.54 ± 1.65	92.69 ± 1.11	95.38 ± 0.77	**96.12 ± 0.62**

TABLE III

CLASSIFICATION EXPERIMENT OF RESULTS ON THE PU (%). BOLD INDICATES THE BEST RESULTS. STANDARD DEVIATIONS ARE APPENDED

Class	2D-CNN	Spectral Former	SSGRN	DMVL + SVM	S3Net	GGCL	SSCL	ConVaT	FTCL
1	67.82 ± 3.45	61.63 ± 4.12	77.23 ± 3.15	78.64 ± 2.86	90.30 ± 1.12	91.17 ± 1.05	90.24 ± 1.23	95.86 ± 0.76	**96.86 ± 0.49**
2	80.87 ± 2.78	86.89 ± 2.14	92.44 ± 1.07	92.58 ± 1.05	97.15 ± 0.55	96.41 ± 0.66	97.15 ± 0.51	96.80 ± 0.60	**98.22 ± 0.25**
3	57.91 ± 4.78	67.27 ± 4.31	52.44 ± 5.23	63.98 ± 4.01	63.72 ± 4.03	79.39 ± 2.44	**93.77 ± 1.27**	70.33 ± 3.95	87.80 ± 1.71
4	36.80 ± 6.34	32.44 ± 6.91	87.22 ± 2.89	79.57 ± 3.36	93.87 ± 1.03	87.32 ± 1.97	95.97 ± 1.04	96.09 ± 0.88	**96.15 ± 0.49**
5	57.65 ± 4.74	84.16 ± 2.93	99.13 ± 0.12	96.54 ± 0.48	93.89 ± 1.08	**99.54 ± 0.05**	98.23 ± 0.43	97.30 ± 0.66	95.26 ± 0.90
6	93.46 ± 1.26	98.60 ± 0.12	77.44 ± 3.17	95.36 ± 0.60	96.98 ± 0.97	98.33 ± 0.15	**99.62 ± 0.03**	97.89 ± 0.57	98.06 ± 0.42
7	**81.76 ± 2.51**	93.70 ± 1.55	92.26 ± 1.67	86.06 ± 2.58	79.33 ± 2.74	94.44 ± 0.98	88.03 ± 1.94	**98.28 ± 0.38**	94.80 ± 1.19
8	60.01 ± 4.69	60.37 ± 4.47	59.62 ± 4.43	74.53 ± 3.24	72.77 ± 3.18	80.31 ± 2.19	74.74 ± 3.30	81.71 ± 2.74	**89.55 ± 1.17**
9	48.91 ± 5.78	80.05 ± 3.73	74.88 ± 3.55	79.35 ± 3.26	81.49 ± 2.83	81.65 ± 2.56	85.79 ± 1.91	82.81 ± 2.77	**94.09 ± 1.10**
OA	74.98 ± 2.63	76.85 ± 2.44	82.58 ± 1.84	86.87 ± 1.56	90.85 ± 1.11	92.71 ± 0.92	93.39 ± 0.80	94.47 ± 0.62	**95.55 ± 0.40**
AA	55.88 ± 4.35	64.31 ± 3.97	69.27 ± 3.82	77.95 ± 2.99	83.87 ± 2.34	87.26 ± 1.72	88.82 ± 1.28	90.29 ± 1.01	**93.03 ± 0.53**
Kappa	65.08 ± 3.54	68.68 ± 3.28	76.64 ± 2.88	82.31 ± 2.44	87.82 ± 1.73	90.26 ± 1.19	91.19 ± 1.09	92.65 ± 0.70	**94.07 ± 0.55**

TABLE IV

CLASSIFICATION EXPERIMENT OF RESULTS ON THE LONGKOU (%). BOLD INDICATES THE BEST RESULTS. STANDARD DEVIATIONS ARE APPENDED

Class	2D-CNN	Spectral Former	SSGRN	DMVL + SVM	S3Net	GGCL	SSCL	ConVaT	FTCL
1	60.52 ± 5.20	77.09 ± 4.12	89.60 ± 2.84	95.02 ± 1.45	99.85 ± 0.10	98.84 ± 0.21	94.43 ± 1.66	**100.0 ± 0.00**	99.85 ± 0.10
2	83.96 ± 3.11	90.91 ± 2.54	99.08 ± 0.42	99.54 ± 0.25	98.15 ± 0.55	87.91 ± 3.21	99.36 ± 0.28	98.49 ± 0.52	**100.0 ± 0.00**
3	88.76 ± 2.42	50.01 ± 5.87	99.95 ± 0.05	98.03 ± 0.79	88.06 ± 2.92	99.62 ± 0.11	**100.0 ± 0.00**	88.30 ± 3.02	**100.0 ± 0.00**
4	90.95 ± 2.48	89.68 ± 2.70	90.00 ± 2.97	98.89 ± 0.54	95.21 ± 1.81	90.62 ± 2.14	93.27 ± 1.67	95.35 ± 1.17	**99.20 ± 0.20**
5	97.04 ± 0.87	78.56 ± 3.65	95.43 ± 1.43	88.74 ± 3.58	98.71 ± 0.43	91.02 ± 2.74	84.79 ± 4.13	98.89 ± 0.55	**99.07 ± 0.31**
6	75.20 ± 3.78	**100.0 ± 0.00**	98.08 ± 0.50	93.31 ± 2.69	97.84 ± 0.51	99.90 ± 0.05	99.80 ± 0.04	94.04 ± 1.44	99.48 ± 0.11
7	**100.0 ± 0.00**	70.27 ± 5.02	99.00 ± 0.48	97.43 ± 1.23	**100.0 ± 0.00**	97.83 ± 0.63	98.97 ± 0.28	**100.0 ± 0.00**	**100.0 ± 0.00**
8	91.59 ± 2.87	72.30 ± 4.27	86.06 ± 3.79	95.75 ± 1.38	93.10 ± 2.37	97.81 ± 0.34	93.92 ± 1.70	**98.16 ± 0.41**	95.06 ± 1.25
9	98.16 ± 0.67	91.65 ± 2.35	94.48 ± 1.59	**100.0 ± 0.00**	94.34 ± 1.44	97.43 ± 0.47	**100.0 ± 0.00**	99.53 ± 0.11	**100.0 ± 0.00**
OA	85.92 ± 2.61	76.61 ± 3.12	93.86 ± 1.26	95.37 ± 1.14	94.09 ± 1.11	95.46 ± 1.01	95.33 ± 0.99	96.41 ± 0.78	**97.35 ± 0.44**
AA	77.86 ± 3.01	76.82 ± 3.27	95.29 ± 0.94	94.68 ± 1.15	95.58 ± 0.79	94.43 ± 0.88	94.79 ± 0.82	96.79 ± 0.62	**97.59 ± 0.32**
Kappa	84.32 ± 2.58	73.98 ± 3.41	93.15 ± 1.22	94.85 ± 1.08	93.43 ± 1.10	94.95 ± 0.97	94.81 ± 0.86	95.99 ± 0.69	**97.05 ± 0.37**

the use of unlabeled data, reducing reliance on labeled samples. The combination of effective data augmentation and asymmetric feature extraction significantly improves classification performance, particularly in scenarios with limited labeled data.

2) *MFNL Module:* The MFNL module models the ambiguity of multiview samples through fuzzy membership calculation and weighted feature aggregation. By integrating spatial contextual information, it extracts robust spectral–spatial features while suppressing interference from neighboring pixels.

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TABLE V
CLASSIFICATION EXPERIMENT OF RESULTS ON THE HANCHUAN (%). BOLD INDICATES THE BEST RESULTS. STANDARD DEVIATIONS ARE APPENDED

Class	2D-CNN	Spectral Former	SSGRN	DMVL + SVM	S3Net	GGCL	SSCL	ConVaT	FTCL
1	**100.0 ± 0.00**	**100.0 ± 0.00**	85.39 ± 3.50	**100.0 ± 0.00**	83.78 ± 3.27	92.18 ± 2.11	**100.0 ± 0.00**	96.97 ± 1.10	**100.0 ± 0.00**
2	83.67 ± 3.24	87.19 ± 2.88	95.52 ± 1.22	93.19 ± 1.67	91.25 ± 2.10	91.30 ± 1.95	90.64 ± 1.76	93.93 ± 1.50	**97.24 ± 0.70**
3	85.80 ± 3.02	92.89 ± 1.95	79.86 ± 3.64	94.46 ± 1.54	80.87 ± 3.12	95.14 ± 1.11	**98.48 ± 0.52**	94.58 ± 1.37	94.55 ± 0.84
4	92.63 ± 2.30	85.52 ± 2.90	97.67 ± 0.96	93.85 ± 1.14	98.14 ± 0.88	93.08 ± 1.78	95.39 ± 1.12	98.30 ± 0.66	**98.55 ± 0.28**
5	91.94 ± 2.45	78.81 ± 3.38	97.74 ± 0.90	**100.0 ± 0.00**	95.08 ± 1.15	98.73 ± 0.45	97.82 ± 0.62	91.69 ± 2.17	92.59 ± 1.86
6	99.14 ± 0.38	95.06 ± 1.41	73.57 ± 3.93	98.54 ± 0.70	91.67 ± 2.18	89.76 ± 2.26	98.56 ± 0.52	99.57 ± 0.19	**99.84 ± 0.06**
7	**100.0 ± 0.00**	**100.0 ± 0.00**	91.68 ± 1.25	94.34 ± 1.53	95.15 ± 1.01	79.73 ± 3.79	**100.0 ± 0.00**	**100.0 ± 0.00**	**100.0 ± 0.00**
8	97.77 ± 0.69	98.36 ± 0.52	95.57 ± 0.86	**100.0 ± 0.00**	**100.0 ± 0.00**	99.44 ± 0.15	**100.0 ± 0.00**	**100.0 ± 0.00**	**100.0 ± 0.00**
9	**100.0 ± 0.00**	**100.0 ± 0.00**	75.52 ± 3.82	74.19 ± 3.71	87.70 ± 3.45	94.96 ± 1.53	**100.0 ± 0.00**	88.89 ± 3.12	**100.0 ± 0.00**
10	95.12 ± 1.48	89.64 ± 2.41	98.40 ± 0.83	94.51 ± 1.49	**98.79 ± 0.44**	92.71 ± 1.76	87.74 ± 2.23	96.05 ± 1.25	92.19 ± 1.81
11	73.18 ± 4.76	66.35 ± 5.12	**98.52 ± 0.21**	93.46 ± 1.96	91.04 ± 1.84	83.97 ± 3.46	93.60 ± 2.05	91.20 ± 2.21	94.22 ± 1.33
12	89.72 ± 2.19	87.43 ± 2.88	67.62 ± 4.23	87.26 ± 3.12	92.95 ± 2.24	95.19 ± 1.52	93.48 ± 1.89	95.83 ± 1.14	**97.46 ± 0.49**
13	**100.0 ± 0.00**	**100.0 ± 0.00**	96.94 ± 1.11	92.82 ± 1.96	93.93 ± 1.73	97.82 ± 0.64	93.16 ± 1.74	97.47 ± 1.05	**100.0 ± 0.00**
14	99.75 ± 0.20	99.34 ± 0.34	92.94 ± 1.47	91.28 ± 2.19	97.49 ± 0.81	96.28 ± 0.79	99.59 ± 0.23	99.50 ± 0.18	**99.95 ± 0.05**
15	94.66 ± 1.53	92.88 ± 1.92	91.80 ± 1.58	97.50 ± 1.01	91.90 ± 1.74	94.15 ± 1.45	97.47 ± 0.62	94.03 ± 1.72	**99.71 ± 0.12**
16	**100.0 ± 0.00**	90.25 ± 2.12	98.28 ± 0.66	96.48 ± 1.01	96.90 ± 0.77	99.38 ± 0.17	93.59 ± 1.89	97.54 ± 0.83	98.71 ± 0.42
OA	87.98 ± 2.65	84.12 ± 2.85	91.71 ± 1.84	93.95 ± 1.23	92.78 ± 1.48	94.57 ± 1.21	94.81 ± 1.13	95.20 ± 0.91	**96.99 ± 0.43**
AA	93.92 ± 2.11	86.02 ± 2.55	88.78 ± 2.32	93.87 ± 1.43	92.86 ± 1.32	93.24 ± 1.29	96.22 ± 0.94	95.71 ± 0.86	**97.54 ± 0.39**
Kappa	86.39 ± 2.60	81.89 ± 2.91	90.33 ± 1.76	93.26 ± 1.27	91.94 ± 1.48	93.93 ± 1.06	94.08 ± 1.14	94.55 ± 0.82	**95.41 ± 0.44**

This module enhances feature discriminability and effectively addresses pixel uncertainty, particularly in complex HSI data.

3) *TFD Classifier:* The TFD classifier optimizes feature representation by improving intraclass compactness and interclass separability. By constructing triplets (anchor, positive, and negative samples), it minimizes distances within the same class and maximizes distances between different classes. This structured modeling ensures precise and reliable classification results.

2) Visualization: To enhance the visual interpretability of classification results, Figs. 5–8 present the classification performance across four datasets. Comprehensive analysis reveals that FTCL outperforms all comparison methods on each dataset, primarily due to the synergy of its multiview asymmetric CL framework, MFNL module, and TFD classifier. On the IP dataset, FTCL demonstrates outstanding performance on the SoybeanMintill class, which has extremely limited training samples. The classification boundaries are clear, and spectral discrimination is strong. This is attributed to the CL framework, which enhances interclass separability while maintaining intraclass compactness through the triplet view space, effectively addressing the challenges of small-sample scenarios. Additionally, the MFNL module reduces noise interference, further improving the clarity of classification boundaries. On the PU dataset, FTCL achieves clearer classification boundaries and higher robustness under high-dimensional data characteristics, particularly avoiding misclassification in the BareSoil class, where other methods fail. This improvement is due to the MFNL module's precise modeling of the fuzziness in multiview samples, which enhances the stability of feature representations. On the LongKou dataset, FTCL performs exceptionally well on the Water and Reed classes, with classification results highly consistent with actual ground conditions. However, for spectrally similar classes such as BareLand and BuiltUp, the classification results show slight confusion, likely due to the CL framework's inability to fully distinguish these classes through generated positive and negative sample pairs. On the HanChuan dataset, FTCL excels in the PaddyField and Woodland classes, producing classification results with high spatial integrity and smoothness. This is attributed to the TFD classifier, which optimizes intraclass compactness and interclass separability. However, FTCL performs slightly worse than ConVaT on the BuiltUp class, likely due to the complex spectral characteristics of this class and its overlap with neighboring classes.

In summary, FTCL demonstrates superior classification performance and robustness in HSI classification tasks through its innovative modules, particularly excelling in complex scenarios and small-sample conditions.

D. Effect of Patch Sizes

To assess the influence of patch sizes on classification performance, a range of patch sizes [7, 9, 11, 13, 15] was evaluated across four datasets, using OA as the primary metric. As shown in Fig. 9, the classification performance of the proposed method varies with different patch sizes. Larger patch sizes capture more contextual and neighboring information, thereby improving feature representation capabilities. However, they also increase the complexity of the feature extraction process. Among the tested sizes, a patch size of 13×13 yielded the highest classification accuracy across all datasets. Therefore, 13×13 was selected as the optimal patch size for subsequent experiments, achieving a balance between enhanced classification performance and computational efficiency.

E. Effect of Batch Sizes

Batch size is a critical factor in model training, as larger batch sizes typically require greater hardware resources and

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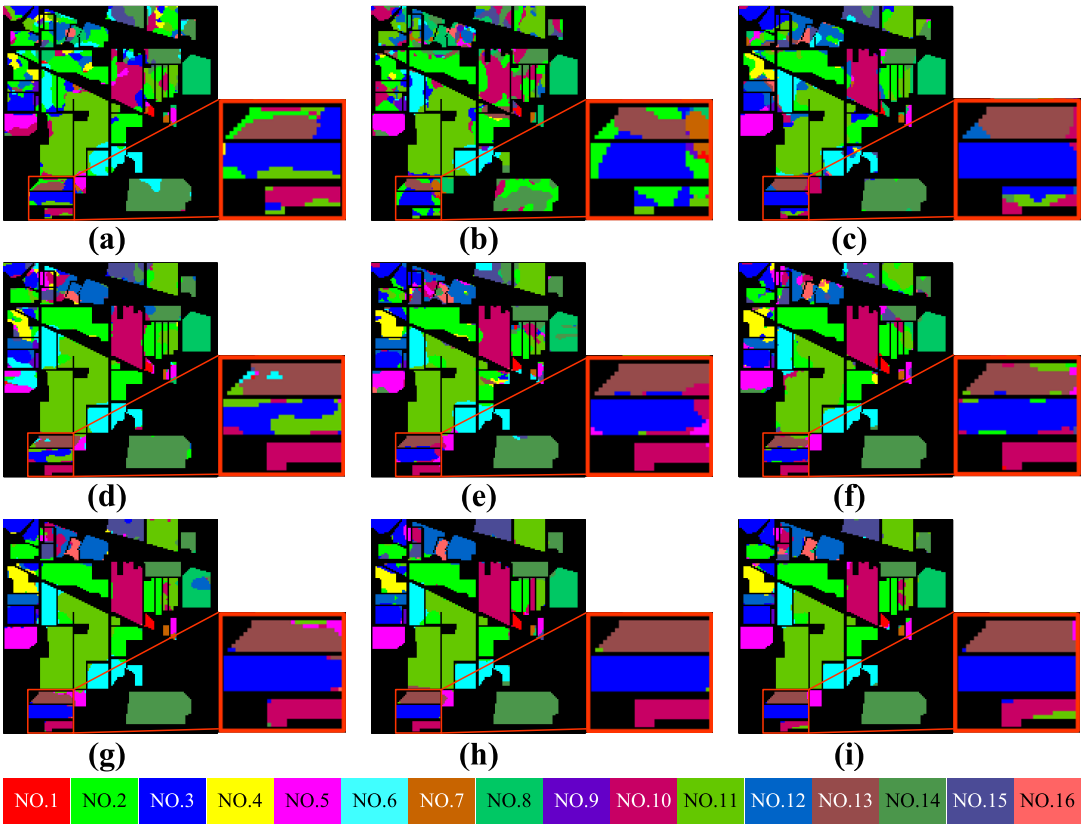


Fig. 5. Classification maps of each algorithm on the Indian Pines dataset. (a) 2D-CNN (65.98%). (b) Spectral Former (57.99%). (c) SSGRN (79.93%). (d) DMVL+SVM (83.41%). (e) S3Net (84.55%). (f) GGCL (87.26%). (g) SSCL (93.61%). (h) ConVaT (95.95%). (i) FTCL (96.60%).

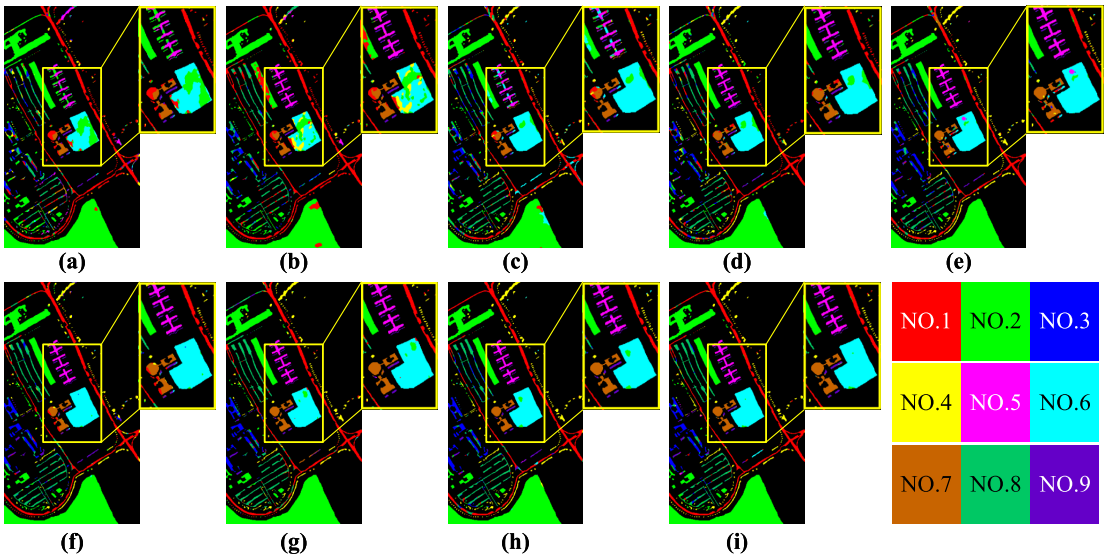


Fig. 6. Classification maps of each algorithm on the PU dataset. (a) 2D-CNN (74.98%). (b) Spectral Former (76.85%). (c) SSGRN (82.58%). (d) DMVL+SVM (86.87%). (e) S3Net (90.85%). (f) GGCL (92.71%). (g) SSCL (93.39%). (h) ConVaT (94.47%). (i) FTCL (95.55%).

increase the complexity of the learning process. In this study, we investigated the impact of batch sizes on the classification performance of FTCL across four datasets to identify the optimal configuration. As shown in Fig. 10, a batch size of 64 achieved the best classification performance on four datasets. Smaller batch sizes tend to make the model more susceptible to noise, resulting in inaccurate gradient estimates, which can destabilize the optimization process and lead to

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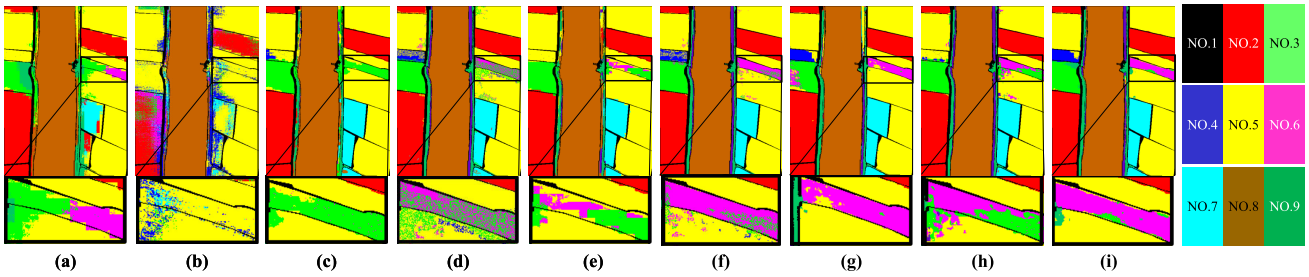


Fig. 7. Classification maps of each algorithm on the LongKou dataset. (a) 2D-CNN (85.92%). (b) Spectral Former (76.61%). (c) SSGRN (93.86%). (d) DMVL+SVM (95.37%). (e) S3Net (94.09%). (f) GGCL (95.46%). (g) SSCL (95.33%). (h) ConVaT (96.41%). (i) FTCL (97.35%).

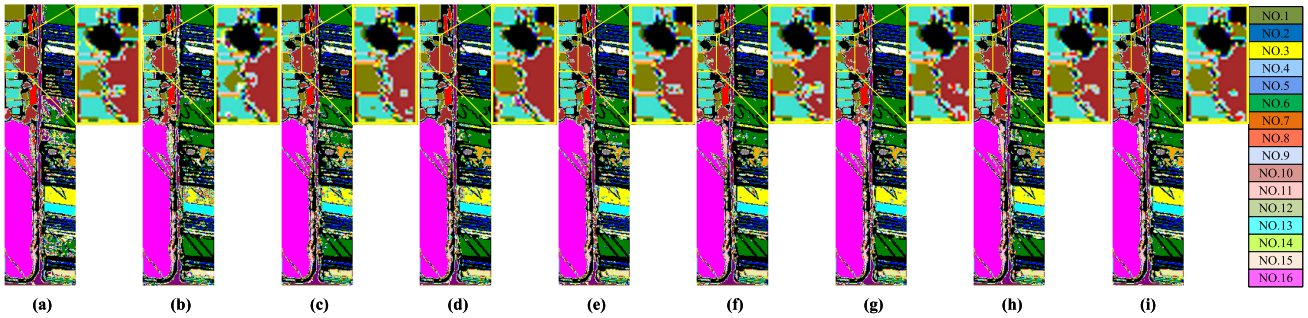


Fig. 8. Classification maps of each algorithm on the HanChuan dataset. (a) 2D-CNN (87.98%). (b) Spectral Former (84.12%). (c) SSGRN (91.71%). (d) DMVL+SVM (93.95%). (e) S3Net (92.78%). (f) GGCL (94.57%). (g) SSCL (94.81%). (h) ConVaT (95.20%). (i) FTCL (96.99%).

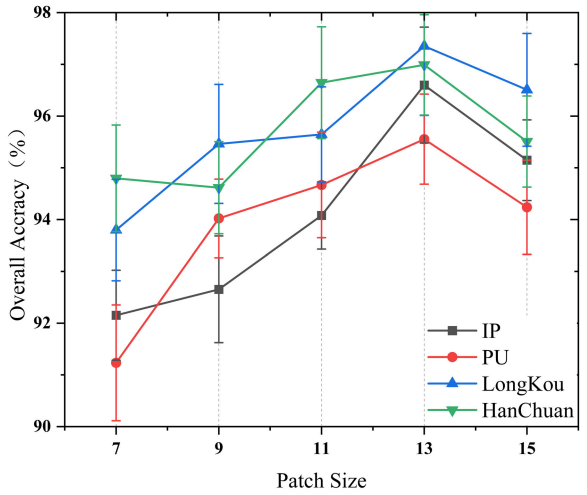


Fig. 9. Effect of patch sizes on the OA metric.

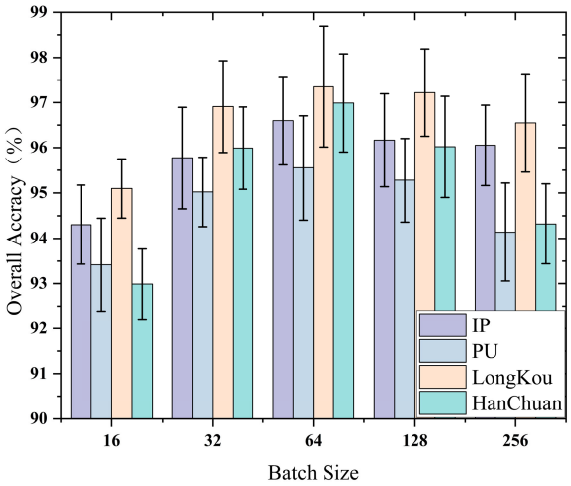


Fig. 10. Effect of batch sizes on the OA metrics.

an unstable gradient descent. In contrast, excessively large batch sizes may cause issues such as vanishing or exploding gradients. Based on the experimental results, a batch size of 64 was determined to be the optimal choice, offering a balance between classification performance and training stability.

F. Comparison of Computational Efficiency

In the quantitative evaluation, we analyze both the accuracy and efficiency of FTCL. Table VI compares the training and testing times, parameter counts, and computational complexity of various methods across four benchmark datasets. Due to their simple architectures, 2D-CNN and Spectral Former exhibit low time and resource consumption, with 2D-CNN leveraging basic convolutional operations and Spectral Former utilizing efficient Transformer encoders. In contrast, SSGRN and DMVL+SVM require more time and have lower complexity and parameter counts, but their feature representation capacity is limited. Two-stage CL methods, such as S3Net, GGCL, SSCL, and ConVaT, are time-consuming and have higher Params and GFLOPs, but they achieve better classification performance.

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FTCL, while requiring a relatively longer training time, maintains a moderate parameter count and computational complexity compared to other advanced methods. This efficient model design enables FTCL to effectively integrate both CL and fuzzy neighborhood strategies without incurring excessive resource overhead. As a result, FTCL delivers superior classification accuracy while keeping inference time and memory consumption competitive with mainstream approaches. The balance between increased representational capacity and manageable computational cost demonstrates that FTCL is well-suited for real-world HSI classification tasks, where both effectiveness and efficiency are essential.

G. Ablation Analysis

To evaluate the contributions of each module in FTCL, we conducted ablation studies, with results summarized in Table VII. The experiments are structured as follows: Experiment 1 uses the original BYOL as the baseline; Experiment 2 integrates the MFNL module for fuzzy feature processing; Experiment 3 introduces negative samples and the TFD classifier for auxiliary constraints; and Experiment 4 combines all modules.

The results of Experiment 2 show that the MFNL module significantly improves classification by leveraging spectral neighborhood information to enhance interclass accuracy and reduce edge effects. Its fuzzy clustering approach mitigates noise, such as spectral aliasing and sensor interference, improving model robustness. Experiment 3 demonstrates that adding negative samples and the TFD classifier enhances feature diversity and representation, increasing OA by 2.138%, 3.691%, and 3.231% across datasets. The TFD classifier further strengthens interclass separability and intraclass consistency, making features more discriminative. In Experiment 4, combining all modules improves OA by 4.013%, 4.941%, and 3.762% compared to the baseline. The TFD classifier module and negative samples capture global features and reduce pixel misclassification, while the MFNL module aggregates spatial contextual information, enhancing interclass accuracy. In summary, the TFD classifier and negative samples improve global feature learning and classification accuracy, while the MFNL module enhances local robustness. Their collaboration enables FTCL to achieve significant performance gains in classification tasks.

Fig. 11 presents the t-SNE-based feature visualization of the ablation experiment results on the PU dataset. As the modules are progressively integrated, the feature discriminability improves significantly, with intraclass clusters becoming more compact and interclass distributions more distinctly separated. This demonstrates that the synergistic effect of the modules substantially enhances the discriminative capability of the features.

H. Effect of Label Amount

To further evaluate the classification performance of FTCL under varying sample size conditions, we conducted experiments on the IP, PU, LongKou, and HanChuan datasets with sample sizes ranging from 10 to 60. As shown in Fig. 12, FTCL consistently achieved excellent performance and exhibited a stable upward trend as the sample size increased. Notably, FTCL demonstrated strong robustness and sensitivity to small training sample sizes, maintaining superior accuracy even when the sample size was as low as 10. This indicates that FTCL effectively leverages limited labeled data to extract discriminative features, outperforming ConVaT, SSCL, GGCL, and S3Net, which, while also showing an upward trend, achieved lower overall performance. These results underscore the stability, adaptability, and superiority of FTCL, particularly in scenarios with limited training samples.

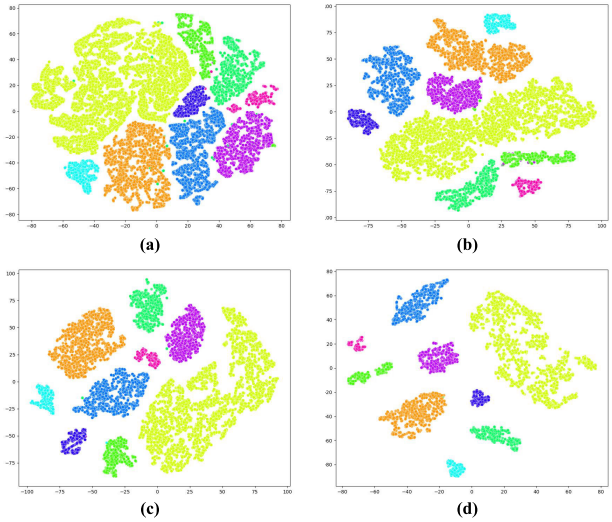


Fig. 11. Feature visualization of ablation experiment results on the PU dataset by t-SNE. (a) Baseline model. (b) MFNL+ BYOL. (c) TFD classifier + BYOL. (d) FTCL.

I. Statistical Analysis

To assess the experimental results obtained using 9 methods on the IP, PU, LongKou, and HanChuan datasets, it is imperative to not only compare and analyze their performance but also to evaluate if there are significant differences overall. The Friedman test is utilized to analyze the ranking of data and ascertain whether there are notable discrepancies among the experimental results from various methods. Initially, the null hypothesis is established, assuming that the results of all experimental methods exhibit the same distribution and no significant differences. The test statistics χ_F^2 and F_F are calculated using the following formula:

$$\chi_F^2 = \frac{12N}{k(k+1)}\left[\sum_{i=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right] \tag{26}$$

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2} \tag{27}$$

where R_j represents the total ranking of the j_{th} method, k represents the number of methods used in the experiment, and N represents the number of datasets used in the experiment. In this study, $k = 9$ and $N = 4$.

Under the condition of $\alpha = 0.05$ significance level of 0.005, we calculated the statistical results of OA, AA, and Kappa for FTCL and other comparative methods, as shown in Table VIII.

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TABLE VI
COMPARISON OF COMPUTATIONAL EFFICIENCY AMONG DIFFERENT MODELS

Method	IP		PU		LongKou		HanChuan		Params	Complexity
	Train	Test	Train	Test	Train	Test	Train	Test	(K)	(GFLOPs)
2D-CNN	31.963	1.201	74.983	5.127	85.436	5.781	85.678	5.678	207.6	0.121
Spectral Former	42.146	1.229	83.852	5.190	76.606	5.778	78.901	5.456	319.4	0.639
SSGRN	37.008	1.206	90.583	5.121	92.187	5.788	88.456	5.789	186.9	0.101
DMVL+SVM	83.408	1.373	81.867	5.126	87.914	5.826	89.123	5.901	126.7	0.092
S3Net	84.550	1.169	150.197	5.087	39.174	5.862	90.789	6.123	438.2	1.032
GGCL	38.927	1.682	124.081	5.083	88.203	5.776	92.345	6.456	271.8	0.612
SSCL	90.199	1.572	155.156	5.167	78.332	5.890	94.123	6.789	198.3	0.478
ConVaT	111.436	1.551	172.324	5.085	98.019	5.955	95.678	7.012	317.5	1.129
FTCL	127.385	1.441	151.837	5.135	109.406	6.067	97.890	7.345	329.7	1.318

TABLE VII
ANALYSIS OF ABLATION EXPERIMENTS

	TFD classifier	MFNL	BYOL	IP		PU		LongKou		HanChuan	
				OA	AA	OA	AA	OA	AA	OA	AA
Experiment1	—	—	✓	92.585	88.778	90.613	82.279	93.590	89.636	92.300	88.100
Experiment2	—	✓	✓	93.587	89.781	92.614	86.283	94.588	91.635	94.145	90.950
Experiment3	✓	—	✓	94.723	92.593	94.304	92.972	96.821	94.596	95.778	94.230
Experiment4	✓	✓	✓	**96.598**	**94.222**	**95.554**	**93.033**	**97.352**	**97.590**	**96.990**	**97.540**

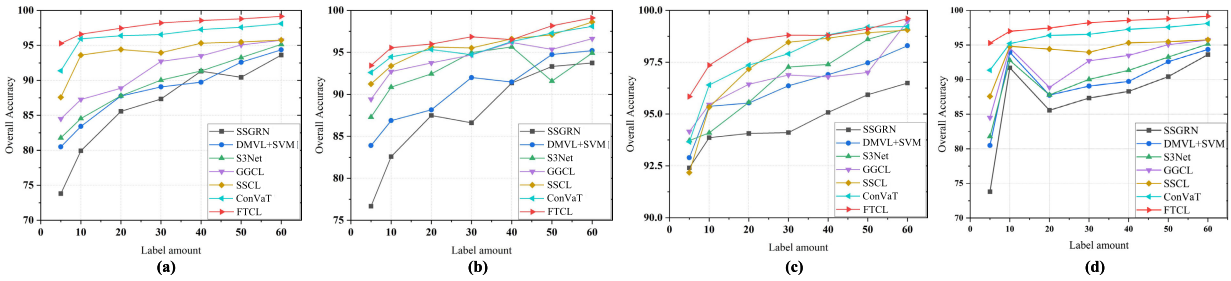


Fig. 12. OA using different amounts of labeled data per class. (a) OAs on the Indian Pines dataset. (b) OAs on the PU dataset. (c) OAs on the LongKou dataset. (d) OAs on the HanChuan dataset.

By calculation, the critical value of the F distribution is $F_\alpha(N,k) = F_{0.05}(4,9) = 3.86$. Table VIII shows that all three indicators have values exceeding 3.86, leading to the rejection of the null hypothesis. This suggests significant differences in the experimental results obtained from various methods. Following nonparametric statistical analysis, we employed the Nemenyi test for multiple group comparisons to further investigate the extent of differences among the methods. The critical difference (CD) of the statistical indicators was calculated using the formula

$$\mathrm{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}} \tag{28}$$

where the significance level $\alpha = 0.05$, $q_{0.05} = 3.031$ is the corresponding lookup constant, and $\mathrm{CD} = 6.384$ is calculated from this. For each pair of comparison methods, the average rank difference is calculated. If the difference exceeds the critical range of 6.384, it is considered that there is a significant difference between these two methods.

Fig. 13 illustrates the average rank differences of various methods across three evaluation metrics. ConVat, SSCL, and GGCL show strong competitiveness in these metrics, surpassing S3Net, DMVL+SVM, and SSGRN. While other comparison methods also display competitiveness in classification results, the FTCL method outperforms them across all three evaluation metrics.

TABLE VIII
STATISTICAL RESULTS OF THE THREE EVALUATION METRICS FOR DIFFERENT METHODS AT A SIGNIFICANCE LEVEL OF $\alpha = 0.05$

	OA	AA	Kappa
χ_F^2	17.32	17.11	17.32
F_F	19.68	18.51	19.68

V. DISCUSSION

Existing CL-based HSI classification methods have the following two shortcomings: 1) multiview samples generated through data augmentation introduce ambiguity, making it difficult to achieve robust feature learning; and 2) feature discrepancies between views reduce intraclass compactness and interclass separability, limiting the discriminative capability of traditional CL methods.

To address these challenges, FTCL introduces two key components: the MFNL module and the TFD classifier. The MFNL module mitigates ambiguity in multiview samples through fuzzy membership and weighted aggregation, enhancing feature robustness. The TFD classifier strengthens intraclass compactness and interclass separability by refining

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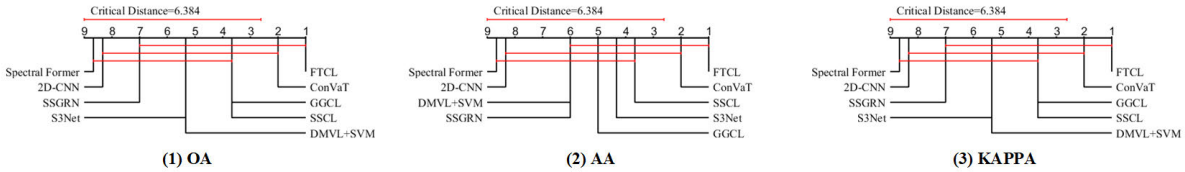


Fig. 13. Nemenyi test results of FTCL compared with the other 8 methods. (a) OA. (b) AA. (c) KAPPA.

anchor–positive–negative relationships, enabling more discriminative feature learning.

To verify the effectiveness of the FTCL, we conducted a series of experiments. In Section IV-C, we conducted experiments on four datasets to compare FTCL with eight different HSI classification methods. The experimental results show that the classification performance of FTCL is significantly better than existing CL-based HSI classification methods. Moreover, in Section IV-G, we further validated the contribution of each module in the FTCL method to the classification performance of the model through ablation experiments.

In fact, FTCL also has the following limitations.

1) *Computational Complexity:* FTCL achieves high classification accuracy but comes with relatively high computational demands. The MFNL module requires calculating fuzzy membership for each pixel in its neighborhood, and the triplet selection process increases computational cost and time consumption, particularly for large-scale HSIs.

2) *Parameter Sensitivity:* FTCL involves several hyperparameters, such as those controlling the fuzzy membership functions, which are relatively sensitive and require careful tuning. This dependency on parameter adjustment may limit the method's ease of use and general applicability across different datasets. Future work will focus on developing more adaptive and robust parameter-tuning mechanisms to alleviate this limitation.

VI. Conclusion

In this study, we propose FTCL, a self-supervised learning framework designed for HSI classification. FTCL addresses key challenges such as multiview ambiguity and feature discrimination by integrating the MFNL module and the TFD classifier. The MFNL module enhances robustness to noise and uncertainty through fuzzy processing, while the TFD classifier improves intraclass compactness and interclass separability, enabling more discriminative feature learning. FTCL employs a two-stage architecture, where pretraining captures spectral–spatial characteristics and strengthens feature discrimination, and fine-tuning generates high-quality features for classification. Experiments on benchmark datasets demonstrate the superior performance of FTCL, particularly with limited labeled samples, validating its effectiveness and robustness.

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